



Forecasting realized volatility using HAR models and wavelet decomposition: A volatility-timing perspective

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ABSTRACT

This study proposes a wavelet-based approach to forecasting Realized Volatility (RV) and evaluates its economic value within a volatility-timing framework. We apply wavelet decomposition to separate short-, medium-, and long-term components and generate forecasts using Heterogeneous Autoregressive (HAR) models. Forecasts based on the low-frequency component consistently lead to better portfolio outcomes, reducing turnover and enhancing investor utility without increasing risk. These results hold even when portfolio weights are forecast directly after being constructed from RV, or when jump-robust volatility estimates are used. The results highlight the importance of aligning forecast evaluation with practical investment objectives. Forecasts delivering the greatest welfare gains may not minimize conventional statistical loss functions.

1. Introduction

Forecasting volatility in equity markets has long been a focus of finance research. Much of the literature builds on [Bollerslev's \(1986\)](#) generalized autoregressive conditional heteroskedasticity (GARCH) model, which models conditional variance as a function of past shocks and variances to capture volatility clustering in asset returns. GARCH variants remain widely used due to their ability to dynamically respond to new information.

A related strand of literature ([Alfeus & Nikitopoulos, 2022](#); [Bonato et al., 2024](#); [Degiannakis et al., 2022](#); [Dutta et al., 2024](#); [Ma et al., 2019](#); [Marfatia et al., 2022](#); [Souropanis & Vivian, 2023](#); [Tian et al., 2017](#); [Zhang & Zhang, 2023](#)) comprises studies that use the heterogeneous autoregressive (HAR) model introduced by [Corsi \(2009\)](#) for forecasting realized volatility. HAR models have several desirable features. They lend themselves well to analyzing high-frequency data. They are parsimonious and easy to interpret, with their simple structure providing a clear view of how current volatility is associated with volatility over the past day, week, and month. Importantly, they yield remarkably accurate forecasts ([Alfeus & Nikitopoulos, 2022](#)). Few studies, however, attempt to improve forecasts by examining the latent components of realized volatility. A notable exception is [Souropanis and Vivian \(2023\)](#), who decomposed volatility using wavelets and found that technical indicators helped at high frequencies while macroeconomic variables mattered at low frequencies.

We contribute to this important and growing body of work by forecasting the realized volatility of the 20 major equities in the S&P 500 and providing recommendations, based on a volatility-timing (VT) strategy, for improving portfolio performance. Our analysis proceeds as follows. First, we estimate a benchmark HAR model to provide a baseline forecast of realized volatility over daily, weekly,

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and monthly horizons. Second, we apply wavelet decomposition to isolate high-, medium-, and low-frequency components of volatility. These components are then forecast separately using HAR-style models. Finally, we evaluate these forecasts through a VT strategy, which uses volatility forecasts to dynamically adjust portfolio weights. As a robustness check, we also generate forecasts using ARMA models and compare their performance with that of the HAR-based forecasts.

Our main contribution lies in leveraging wavelet decomposition to show that some forecasts, while not optimal under standard statistical loss functions, deliver higher economic value through improved portfolio outcomes. We find that the low-frequency component of volatility, though statistically inferior, consistently yields better outcomes for investors. This relationship arises because forecasts based on low-frequency volatility lead to lower portfolio turnover without materially increasing volatility, thereby reducing transaction costs and enhancing investor welfare. This finding reinforces the distinction between statistical accuracy and economic usefulness, and it highlights the importance of evaluating forecasts in a setting aligned with practical portfolio objectives. Reconciling high-, medium-, and low-frequency forecasts adds little in terms of portfolio performance. Using ARMA models to forecast the medium- and high-frequency components does not yield improvements in forecasts relative to HAR; in general, HAR models prove superior.

2. Data and methodological approach

We used high-frequency, five-minute intraday price data from LSEG Datascope to estimate daily realized volatility for 20 major equities in the S&P 500 from 31 January 2000 to 31 December 2024, yielding 6,166 daily observations.

2.1. Estimating volatility

The i -th Δ -period return within day t is defined as

$$r_{t,i} = p_{t-1,i\Delta} - p_{t-1,(i-1)\Delta} \quad (1)$$

for $i = 1, 2, \dots, M$, where $p_{t-1,i\Delta}$ is the logarithm of price and M is the number of intraday intervals within a trading day. Given a five-minute frequency, $M = 78$ in this case. One-day RV is expressed as

$$RV_t = \sum_{i=1}^M r_{t,i}^2 \quad (2)$$

and it is a consistent estimator of the unobserved variance as $M \rightarrow \infty$ (Andersen & Bollerslev, 1998).

2.2. Wavelet decomposition

To extract volatility components across different frequencies, we employed the wavelet decomposition approach used by Sourapanis and Vivian (2023). The wavelet decomposition is based on a mother wavelet function, (ψ) , and a father wavelet function, (ϕ) , which capture the high- and low-frequency characteristics of the series, respectively. Formally,

$$\psi_{j,k}(t) = \frac{\psi(2^{-j}t - k)}{\sqrt{2^j}} \quad (3)$$

and

$$\phi_{j,k}(t) = \frac{\phi(2^{-j}t - k)}{\sqrt{2^j}}, \quad (4)$$

where J is the maximum level of the decomposition, $j = 1, 2, \dots, J$, with a lower value of J representing higher-frequency components, and k , counting from 1 to the number of wavelet coefficients, is the dilation parameter. The multiscale framework enables the separation of short-term and long-term movements in the volatility process, which is important for identifying frequency-specific dynamics.

The wavelet coefficients associated with an arbitrary time series y_t , at scale j and position k , are defined as

$$s_{j,k} = \int \psi_{j,k} y_t dt \quad (5)$$

and

$$d_{j,k} = \int \phi_{j,k} y_t dt \quad (6)$$

here, $s_{j,k}$ and $d_{j,k}$ represent the low-frequency and high-frequency coefficients, respectively.

The Multi-Resolution Analysis (MRA) representation of y_t is given by

$$y_t = \sum_k s_{j,k} \phi_{j,k}(t) + \sum_k d_{j,k} \psi_{j,k}(t) + \sum_k d_{j-1,k} \psi_{j-1,k}(t) + \cdots + \sum_k d_{1,k} \psi_{1,k}(t). \quad (7)$$

This is equivalent to

$$y_t = y_t^{S_j} + \sum_j y_t^{L_j} \quad (8)$$

where $y_t^{S_j} = \sum_k s_{j,k} \phi_{j,k}(t)$ is the smooth component and $y_t^{L_j} = \sum_k d_{j,k} \psi_{j,k}(t)$ are the remaining wavelet components.

We employed the Maximum Overlap Discrete Wavelet Transform (MODWT) with a Haar wavelet filter under reflecting boundary conditions (Souropanis & Vivian, 2023) and decomposed daily RV_t into a smooth component $y_t^{S_j}$ and six wavelet components $y_t^{L_j}$, $j = 1, 2, \dots, 6$. The decomposition of RV_t can be represented as

$$RV_t = RV_t^{L_1} + RV_t^{L_2} + RV_t^{L_3} + RV_t^{L_4} + RV_t^{L_5} + RV_t^{L_6} + RV_t^{S_6} \quad (9)$$

This formulation enabled us to isolate and analyze the contribution of volatility across distinct frequency bands, moving from high- to low-frequency components from left to right.

For the subsequent analysis, three aggregated components were used:

$$RV_t^H = RV_t^{L_1} + RV_t^{L_2} \quad (10)$$

$$RV_t^M = RV_t^{L_3} + RV_t^{L_4}, \quad (11)$$

and

$$RV_t^L = RV_t^{L_5} + RV_t^{L_6}, \quad (12)$$

representing high (H), medium (M), and low (L) frequency components of volatility, respectively.

2.3. Forecasting volatility

A rolling window of 1000 trading days is used for estimation. At each step, we execute the wavelet decomposition and estimate model parameters using only observations available up to that point in time. This design prevents data leakage by ensuring that no future information is used when generating forecasts and is consistent with Degiannakis and Kafousaki (2025). Despite its simplicity, the HAR model effectively captures the long-memory and persistence typically observed in financial volatility series. It is based on lagged moving averages (daily, weekly, and monthly) of volatility and is defined as

$$RV_t = \beta_0 + \beta_1 RV_{t-1}^d + \beta_2 RV_{t-1}^w + \beta_3 RV_{t-1}^m + \varepsilon_t, \quad (13)$$

where

$$RV_{t-1}^d = RV_{t-1}, RV_{t-1}^w = \frac{1}{5} \sum_{i=1}^5 RV_{t-i}, \text{ and } RV_{t-1}^m = \frac{1}{22} \sum_{i=1}^{22} RV_{t-i} \quad (14)$$

Using the estimated coefficients from Eq. (13), volatility forecasts at horizon h were obtained as

$$F_{t+h} = \widehat{\beta}_0 + \widehat{\beta}_1 RV_t^d + \widehat{\beta}_2 RV_t^w + \widehat{\beta}_3 RV_t^m \quad (15)$$

This forecast will simply be denoted as HAR in the subsequent analysis. Longer horizon forecasts ($h > 1$) are generated by replacing RV_t in Eq. (13) with an average over $t \rightarrow t + h$.

Forecasts of the three wavelet components, RV_t^L , RV_t^M , and RV_t^H were constructed in the same manner as Equations (13), (14), and (15). These forecasts will be denoted as HAR-L, HAR-M, and HAR-H, respectively.

To provide forecasts of total RV_t volatility, a naïve approach would simply be to sum the forecasts of the three components. However, this approach ignores the information in the dynamics of the full RV_t series. A hierarchical forecasting structure was taken with RV_t at the top level, and RV_t^L , RV_t^M , and RV_t^H at the lower level. Given the forecasts of the three wavelet components, RV_t^L , RV_t^M , and RV_t^H , the MinT forecast reconciliation approach of Wickramasuriya et al. (2019) was used to provide a reconciled forecast of RV_t at the top level. This ensured the forecasts were coherent across both levels in the hierarchy. The reconciled forecast of total volatility is denoted HAR-REC (Caporin & Costola, 2022).

In addition to HAR-REC, the reconciled forecast of total RV_t , forecasts from the individual wavelet components were used. The forecast of the low-frequency component, denoted HAR-L, was used for volatility, along with the combination of the low- and medium-frequency forecasts, HAR-L and HAR-M, denoted HAR-LM.

2.4. Volatility timing and forecast evaluation

The VT framework developed by Kirby and Ostdiek (2012) was employed to assess the economic value of the forecasting methods proposed in this study, including those based on wavelet-decomposed components. This design reduces estimation error and portfolio turnover, thereby minimizing transaction costs. The framework imposes a strong shrinkage assumption on the traditional mean–variance optimization by treating the off-diagonal elements of the covariance matrix as zero, effectively relying only on univariate volatility forecasts.

Recent studies have emphasized the importance of evaluating volatility forecasts using economic-based criteria rather than relying exclusively on statistical loss functions. For example, Degiannakis and Filis (2022) assessed forecasts through after-cost trading profits from volatility-based trading rules, while Delis et al. (2025) employed trading profitability and risk management loss functions based on Value-at-Risk violations. In the same spirit, the volatility timing framework adopted here evaluates forecasts based on their impact on portfolio allocation, transaction costs, and investor welfare. The analysis here differs from that of Degiannakis and Filis (2022) for instance in that a special case of mean–variance optimization across many assets is considered here, whereas Degiannakis and Filis (2022) considers timing-based strategies on the volatility of a single asset.

Let FC_t , a vector whose elements FC_t^i contain the forecasts of volatility for each equity. The weight allocated to equity i in period t under VT, is defined as

$$w_{i,t} = \frac{(1/FC_t^i)^\eta}{\sum_{i=1}^N (1/F_t^i)^\eta}, \quad (16)$$

where η is a tuning parameter that controls the responsiveness of the weights to changes in the forecasts. Given the portfolio weights, returns are then defined as

$$r_{p,t} = w_t r_t' \quad (17)$$

where w_t is the vector of portfolio weights and r_t is the vector of asset returns. Since portfolio weights respond dynamically to forecast revisions, frequent adjustments may lead to high turnover and, thus, high transaction costs.

A formal portfolio turnover measure is defined as follows. Suppose \$1 is invested in the portfolio at time $t-1$. The value of asset i will become $w_{i,t-1}(1+r_{i,t})$ at time t . The portfolio weight of asset i before rebalancing at time t is given by

$$\widetilde{w}_{i,t} = \frac{1 + w_{i,t}}{1 + \sum_{i=1}^N w_{i,t} r_{i,t}}, \quad (18)$$

with the turnover at time t given by

$$TO_t = \sum_{i=1}^N |w_{i,t+1} - w_{i,t} \widetilde{w}_{i,t}| \quad (19)$$

Portfolio returns after accounting for transaction costs can then be expressed as

$$r_{p,t}^* = w_t r_t' - c TO_t \quad (20)$$

where c is a proportional trading cost.

Following Fleming et al. (2001, 2003) and Bollerslev et al. (2018), a utility-based framework was used to compare the economic benefit of the competing forecasts in the context of VT. This approach moves beyond statistical metrics and directly measures the impact of forecast accuracy on investor welfare.

Assuming a risk-averse investor with quadratic utility and a constant coefficient of risk aversion γ , the utility derived from the returns $r_t^{*,k}$ generated by the forecasts from model k is given by

$$U(r_t^{*,k}, \gamma) = (1 + r_t^{p,k}) - \frac{\gamma}{2(1 + \gamma)} (1 + r_t^{p,k})^2. \quad (21)$$

This utility specification captures the trade-off between return and risk, penalizing large deviations in returns more heavily as the risk aversion of the investor increases.

To compare competing approaches, the economic benefit of using model k over another model j can be determined by solving for Δ_γ in

$$\sum_{t=1}^T U(r_t^{p,k}, \gamma) = \sum_{t=1}^T U(r_t^{p,j} - \Delta_\gamma, \gamma), \quad (22)$$

where Δ_γ represents the maximum an investor is willing to pay to switch from model j to model k . Furthermore, to assess the significance of estimates of Δ_γ and, hence, differences in performance across forecasts, we followed Bollerslev et al. (2018) and used the Reality Check procedure of White (2000). Statistical inference was conducted using the stationary bootstrap proposed by Politis and Romano (1994), with 999 bootstrap samples and an average block length of 22 trading days.

3. Results

3.1. Main results

Table 1 reports results for the 1-day horizon, showing estimates of the difference in economic value, Δ_γ in Eq. (22), between pairs of forecasts. Each estimate reflects gain in utility when switching from the forecast in the row to that in the column, for $\gamma = 1$, given a VT tuning coefficient of $\eta = 2$ and transaction costs of $c = 1\%$ and $c = 2\%$. Positive values of Δ_γ indicate that the forecast model listed in the column heading yields higher investor utility than the one in the row heading. Statistically significant differences at the 5% level based on the Reality Check test are shown in bold.

The results in Table 1 reveal an important result. The low-frequency component RV_t^L proves economically valuable, with the HAR-L model consistently delivering higher utility than HAR. Trying to harness the information contained in the medium-frequency component, RV_t^M , provides no benefits under VT, as HAR-LM underperforms HAR-L and HAR. While often important when evaluating forecasts under statistical loss, forecast reconciliation is not beneficial in the VT framework, with HAR-REC, not outperforming HAR and underperforming HAR-L. This is due to the variability in the forecasts given the high-frequency component is included. The benefit of the HAR-L forecast under VT is an interesting result. HAR-L is clearly an inferior forecast under any standard statistical measure of loss. Still, it performs well under VT, highlighting the distinction between statistical accuracy and economic usefulness, underscoring the importance of aligning forecast evaluation with investor objectives.

Table 2 provides insights into the source of the benefits of the HAR-L forecast. It reports the ratio of the portfolio volatility, σ , and turnover, TO, generated under various forecasts, relative to the benchmark HAR forecast. Results are presented for both 1-day and 5-day investment horizons, and for two values of the volatility sensitivity parameter, η . Across all configurations, HAR-L consistently achieved lower turnover compared to HAR. In terms of volatility σ , HAR-L delivered comparable portfolio volatility relative to HAR. As η increase from $\eta = 2$ to $\eta = 8$, reductions in portfolio volatility were observed.

Importantly, the primary advantage of HAR-L is not in reducing return volatility, but in lowering turnover while preserving volatility levels, thereby mitigating transaction costs. This demonstrates that economic value can be achieved not only through accuracy, but also through efficiency, reinforcing the distinction between statistical and practical performance metrics in volatility forecasting.

The findings for 1-day forecasts with $\eta = 8$, and for 5-day forecasts, using the corresponding values for γ , c , and η from Tables 1 and 2, are reported in Tables 3 and 4, respectively. These confirm the earlier results. Results in Table 3 show that at $\eta = 8$, HAR-LM also begins to provide small significant benefits over HAR. The results in Table 5, for the longer 22-day forecast horizon, also show similar results to the 5-day case. While combining low and medium frequency components in HAR-LM outperformed HAR, HAR-L continued to be superior among all alternatives considered in this study.

Fig. 1 provides an insight into the relative performance of the forecasts over time. Fig. 1 plots the cumulative difference in utility between HAR-L and HAR (top panel), HAR-LM and HAR (middle panel), and HAR-REC and HAR (bottom panel). The top panel indicates that the relative benefit discussed earlier consistently accrued over time. The lower panels show that while the differences in performance for HAR-LM and HAR-REC relative to HAR are small, they consistently underperform. The gaps widen somewhat when the market experiences heightened volatility.

Overall, HAR-L is the preferred forecast in the context of VT, consistently outperforming its competitors across a range of forecast horizons, levels of transactions costs, and volatility sensitivities.

Table 1
Results for the 1-day forecast horizon.

$\gamma = 1, c = 1\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	56.8001	–3.4373	–2.9543
HAR-L	–	–60.2375	–59.7544
HAR – LM	–	–	0.4831
HAR-REC	–	–	–
$\gamma = 1, c = 2\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	114.2270	–6.6731	–5.8601
HAR-L	–	–120.9004	–120.0876
HAR – LM	–	–	0.8130
HAR-REC	–	–	–

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.

Table 2
Portfolio volatility (σ) and turnover (TO) for each forecast.

	HAR-L	HAR-LM	HAR-REC
1-day, $\gamma = 1, c = 1\%, \eta = 2$			
σ	0.9985	1.0162	1.0243
TO	0.1534	1.0472	1.0420
1-day, $\gamma = 1, c = 1\%, \eta = 8$			
σ	0.8920	1.0329	1.0153
TO	0.1617	0.9695	0.9936
5-day, $\gamma = 1, c = 1\%, \eta = 2$			
σ	0.9887	1.0662	1.0027
TO	0.2258	1.0891	1.0601
5-day, $\gamma = 1, c = 1\%, \eta = 8$			
σ	0.7562	1.0686	1.0254
TO	0.2263	0.9990	1.0584

Note: Portfolio volatility (σ) and turnover (TO) for each forecast, scaled relative to the bench HAR forecast. Values less (greater) than 1 indicate lower (higher) volatility or turnover than the HAR forecast.

Table 3
Results for the 1-day forecast horizon.

$\gamma = 1, c = 1\%, \eta = 8$			
	HAR-L	HAR – LM	HAR-REC
HAR	142.5914	3.8952	0.4726
HAR-L	–	–138.6975	–142.1199
HAR – LM	–	–	–3.4226
HAR-REC	–	–	–
$\gamma = 1, c = 2\%, \eta = 8$			
	HAR-L	HAR – LM	HAR-REC
HAR	286.8648	9.0303	1.5024
HAR-L	–	–227.8437	–285.3709
HAR – LM	–	–	–7.5279
HAR-REC	–	–	–

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.

Table 4
Results for the 5-day forecast horizon.

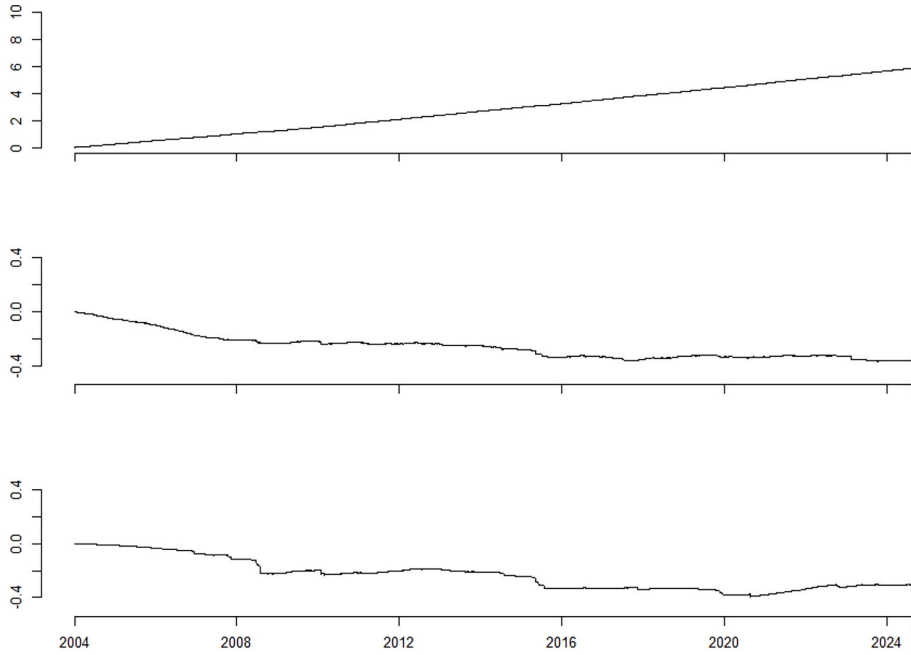
$\gamma = 1, c = 1\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	36.5177	–4.2676	–2.9125
HAR-L	–	–40.7853	–39.4303
HAR – LM	–	–	1.3550
HAR-REC	–	–	–
$\gamma = 1, c = 2\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	73.1998	–8.5252	–5.7646
HAR-L	–	–81.7252	–78.9645
HAR – LM	–	–	2.7606
HAR-REC	–	–	–

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.

Table 5
Results for the 22-day forecast horizon.

$\gamma = 1, c = 1\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	19.3957	15.7199	–0.9650
HAR-L	–	–3.6758	–20.3607
HAR – LM	–	–	–16.6849
HAR-REC	–	–	–
$\gamma = 1, c = 2\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	38.6086	31.2398	–1.8981
HAR-L	–	–7.3688	–40.5067
HAR – LM	–	–	–33.1379
HAR-REC	–	–	–

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.



Note: Cumulative differences in utility between HAR-L and HAR (top panel), HAR-LM and HAR (middle panel), and HAR-REC and HAR (bottom panel). Results are shown for the 1-day forecast horizon.

Fig. 1. Cumulative differences in utility for the 1-day forecast horizon. Note: Cumulative differences in utility between HAR-L and HAR (top panel), HAR-LM and HAR (middle panel), and HAR-REC and HAR (bottom panel). Results are shown for the 1-day forecast horizon.

3.2. Robustness checks: Direct forecasts, jumps, and alternative forecasting models

Golosnoy et al. (2019) and Golosnoy and Gribisch (2022) proposed a direct method for forecasting portfolio weights, which involved constructing a time series of portfolio weights from observed volatility estimates and directly forecasting the observed weights. They showed that the direct forecasting approach leads to improvements in portfolio performance.

Given estimates of the volatility for each asset, RV_t^i , realized portfolio weights can be constructed as

$$w_{it}^* = \frac{(1/RV_t^i)^\eta}{\sum_{i=1}^N (1/RV_t^i)^\eta}. \quad (23)$$

These weights, $w_{t,i}^*$, were treated as individual time series. The equivalent wavelet components (H, M, and L) were constructed in the same manner as outlined earlier. The same HAR framework in Equations (13), (14), and (15) was used to forecast the direct weights and their associated components. The final forecasts of the weights were rescaled to ensure they sum to one. These forecasts of the weights were then used to compute turnover and post-transaction cost returns.

Table 6 presents the results for the 1-day horizon, under the direct forecasting approach, given a VT tuning coefficient of $\eta = 2$ and transaction costs of $c = 1\%$ and $c = 2\%$. Again, entries in bold indicate that the difference in utility is significant at a 5% level of significance according to the Reality Check test.

The results show that HAR-L delivers the highest utility, confirming the effectiveness of incorporating wavelet-based low-frequency components. The benefits of doing so are evident even when volatility forecasts are first transformed into portfolio weights. HAR-REC, while outperforming HAR-LM, underperforms HAR-L. Results for the longer 5- and 22-day direct forecasts are reported in Tables 7 and 8, respectively. They are consistent with the longer-horizon results reported above—HAR-LM performs well at the longer 22-day horizon, but overall, HAR-L continues to be the most effective when using direct-weight forecasts.

Next, we consider the robustness of the results with respect to a jump-robust measure of volatility. Removing the effect of jump activity in prices when estimating volatility, and hence removing the effect of extreme price movements, may be viewed as an alternative to employing low-frequency components of RV. Here we use the continuous component (C_t) of RV as proposed by Andersen et al. (2007). First, define bi-power variation (a jump-robust estimator) as

$$BV_t = \frac{\pi}{2} \left(\frac{M}{M-1} \right) \sum_{i=2}^M |r_{t,i}| |r_{t,i-1}| \quad (24)$$

The continuous component is then defined from the difference between RV_t and BV_t as follows,

$$C_t = I(Z_t \leq \Phi_\alpha) RV_t + I(Z_t > \Phi_\alpha) BV_t, \quad (25)$$

where $I()$ is an indicator function based on a comparison of a jump test-statistic Z_t to a suitable threshold at a significance level of α . Once a series C_t is found, it is used in place of RV_t in all the methodology described earlier.

Table 9 presents results comparing portfolios based on forecasts of C_t , HARC, HARC-L, HARC-LM, and HARC-REC to HAR-L (the best performing approach from the earlier results). Both the HARC and HARC-REC underperform HAR-L, indicating that removing the effects of jumps is not a substitute for using the low-frequency component of RV_t . HARC-L provides equivalent performance to HAR-L, highlighting that the low-frequency component is the dominant effect, and removing jumps provides no further benefits. This is an important result in that one can always apply the wavelet decomposition when RV_t is available, however, the underlying intraday returns are required to compute the jump-robust C_t .

Alternatives to the HAR forecasting model are also considered. The RV_t^M , and RV_t^H components are much less persistent than RV_t^L . As such, the performance of ARMA forecasting models for RV_t^M , and RV_t^H are examined. Automated lag selection is used in every rolling estimation window. Given the very strong persistence of RV_t^L , a HAR model continues to be used. The reconciled forecast of total volatility is denoted as ARMA-R. Tables 10 and 11 present the results for 1- and 5-day forecasts respectively, comparing HAR, HAR-L, HAR-REC and ARMA-R forecasts. At both forecast horizons, HAR-L is superior to the reconciliation-based ARMA-R, a result consistent with the earlier results based on HAR-REC. When comparing the reconciliation-based forecasts, ARMA-R is slightly inferior to HAR-REC at the 1-day horizon and equivalent to HAR-REC at the 5-day horizon. In summary, while the ARMA process offers more flexibility than the fixed lag structure of the HAR, this flexibility does not translate into improved forecasts relative to HAR specifications. In fact, ARMA forecasts are outperformed by HAR forecasts in all but one case, wherein the models performed similarly.

4. Concluding remarks

This paper proposes a wavelet-based decomposition approach to forecast volatility for use in VT strategies. It is found that VT, based on forecasts from the low-frequency component of volatility, leads to improved portfolio outcomes. While maintaining a similar risk profile, forecasts from the low-frequency component of volatility lead to lower portfolio turnover and increases in welfare from the perspective of a risk-averse investor. This result is also observed under the direct forecasting approach, where portfolio weights are constructed from observed volatility and forecast directly. The benefit of the low-frequency component of volatility is still observed when a jump-robust estimate of volatility is used for forecasting and portfolio allocation. Using ARMA models to forecast medium- and higher-frequency volatility components do not improve forecasts relative to HAR-based specifications, with the HAR models proving superior in most cases. These findings underscore that statistical accuracy in volatility forecasts does not necessarily translate into higher economic value; forecasts delivering the greatest welfare gains may not be those that minimize conventional loss functions.

CRedit authorship contribution statement

Adam Clements: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Puneet Vatsa:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization.

Table 6

Direct forecasting results for the 1-day forecast horizon.

$\gamma = 1, c = 1\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	33.9544	–19.9598	–2.4136
HAR-L	–	–53.9142	–36.3680
HAR – LM	–	–	17.5462
HAR-REC	–	–	–
$\gamma = 1, c = 2\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	68.4392	–40.0594	–4.8162
HAR-L	–	–108.4987	–73.3005
HAR – LM	–	–	35.1982
HAR-REC	–	–	–

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.

Table 7

Direct forecasting results for the 5-day forecast horizon.

$\gamma = 1, c = 1\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	14.4068	–15.4458	–0.5976
HAR-L	–	–29.8526	–15.0044
HAR – LM	–	–	14.8483
HAR-REC	–	–	–
$\gamma = 1, c = 2\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	29.0119	–30.8557	–1.2110
HAR-L	–	–59.8676	–30.2229
HAR – LM	–	–	29.6447
HAR-REC	–	–	–

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.

Table 8

Direct forecasting results for the 22-day forecast horizon.

$\gamma = 1, c = 1\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	3.3647	2.2976	–0.3547
HAR-L	–	–1.0670	–3.7193
HAR – LM	–	–	–2.65423
HAR-REC	–	–	–
$\gamma = 1, c = 2\%, \eta = 2$			
	HAR-L	HAR – LM	HAR-REC
HAR	6.7488	4.5891	–0.7094
HAR-L	–	–2.1597	–7.4582
HAR – LM	–	–	–5.2985
HAR-REC	–	–	–

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.

Table 9

Results for the 1-day forecast horizon.

$\gamma = 1, c = 1\%, \eta = 2$				
	HARC	HARC-L	HARC-LM	HARC-REC
HAR-L	-59.4912	0.2642	-60.8038	-61.0312
HARC	—	59.7554	-1.3126	-1.5399
HARC-L	—	—	-61.0680	-61.2954
HARC-LM	—	—	—	-0.2273
HARC-REC	—	—	—	—
$\gamma = 1, c = 2\%, \eta = 2$				
	HARC	HARC-L	HARC-LM	HARC-REC
HAR-L	-119.4523	0.4446	-121.6836	-122.5403
HARC	—	119.8968	-2.2310	-3.0876
HARC-L	—	—	-122.1281	-122.9849
HARC-LM	—	—	—	-0.8566
HARC-REC	—	—	—	—

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.

Table 10

Results for the 1-day forecast horizon.

$\gamma = 1, c = 1\%, \eta = 2$			
	HAR-L	HAR – REC	ARMA-R
HAR	56.8001	-2.9543	-5.7323
HAR-L	—	-59.7544	-62.5324
HAR – REC	—	—	-2.7780
ARMA-R	—	—	—
$\gamma = 1, c = 2\%, \eta = 2$			
	HAR-L	HAR – LM	ARMA-R
HAR	114.2270	-5.8601	-11.3127
HAR-L	—	-120.0876	-125.5400
HAR – REC	—	—	-5.4526
ARMA-R	—	—	—

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.

Table 11

Results for the 5-day forecast horizon.

$\gamma = 1, c = 1\%, \eta = 2$			
	HAR-L	HAR – REC	ARMA-R
HAR	36.5177	-2.9125	0.0055
HAR-L	—	-39.4303	-36.5123
HAR – REC	—	—	2.9180
ARMA-R	—	—	—
$\gamma = 1, c = 2\%, \eta = 2$			
	HAR-L	HAR – LM	ARMA-R
HAR	73.1998	-5.7646	0.0149
HAR-L	—	-78.9645	-73.1849
HAR – REC	—	—	5.7795
ARMA-R	—	—	—

Note: The table reports values of Δ_γ . Forecasts in column headings are compared to those in the row headings. Positive values of Δ_γ indicate the forecast in the column heading is preferred relative to the forecast in the row heading. Bold numbers indicate the difference in utility is significant at a 5% level of significance according to the Reality Check test. Results are shown for $\gamma = 1$ and $\eta = 2$, across increasing levels of transaction costs, $c = 1\%$ and $c = 2\%$.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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